

16S for Red40 Report Stub

Overview

Participants in a controlled study have been consuming diets containing Red 40, a common synthetic food dye. Since some food dyes are processed by the gut bacteria, researchers have collected fecal samples from people with different level of Red 40 exposure and gut bacteria. In this study, there will be two stages: a baseline(pre-exposure) and after a period of Red 40 consumption. This is an important study as synthetic dyes impact the health of many people and alter the microbiome in the gut since specific enzymes will increase to break down food dye.

Data preprocessing

The study aims to investigate if gut microbiome change significantly from baseline to after Red 40 consumption. First, we preprocessed DNA by extracting fecal material using the Macherey-Nagel Stool Kit, following the manufacturer's protocol. The 16S rRNA gene was amplified using universal V1-V9 primers (27F + 1492R). PCR amplification and sample barcoding were simultaneously performed using the Oxford Nanopore SQK-16S024 library preparation kit. DNA concentration was quantified using a Qubit Fluorometer with the Qubit High Sensitivity Assay Kit. Barcoded DNA samples were normalized based on concentration and pooled in equimolar amounts. The pooled library was sequenced across multiple Oxford Nanopore Flongle flowcells (FLO-FLG001) for a total sequencing time of 72 hours. Basecalling was performed using the GUPPY v3.2.2 high-accuracy model. Adapter and barcode trimming were conducted during the basecalling process. Taxonomic classification of the reads was carried out using the EPI2ME 16S workflow (Oxford Nanopore Technologies), which employs reference-based alignment for microbial identification. The resulting classified datasets served as the starting point for downstream microbiome analysis. ## Cleaning Data We were given a dataset of barcodes with multiple fastq files containing raw sequence data for bacteria. For each fastq file, we converted it to fasta files which made it possible to be searched through the BLAST database. Using the BLAST database, we determined the bacteria's taxonomic and OTU data for further analysis.

```
## Set the base image to Ubuntu 20.04 LTS
##FROM --platform=linux/amd64 ubuntu:jammy

## My authorship
## LABEL maintainer="Johnny Chen"
## LABEL version="0.1.0"
## LABEL description="docker_project"

## Disable prompts during package installation
## ENV DEBIAN_FRONTEND=noninteractive

## RUN apt update
## RUN apt install -y curl git g++ wget nano zlib1g-dev make bsdmainutils gawk bcftools libopenblas-base

## Install Miniconda package manager
## RUN cd tmp
## RUN curl https://repo.anaconda.com/miniconda/Miniconda3-py310_23.3.1-0-Linux-x86_64.sh --output minic
```

```

# RUN bash miniconda.sh -bu
# ENV PATH="/root/miniconda3/bin:$PATH"
# RUN conda update -y conda

# # Update conda and install bioconda packages
# RUN conda update -y conda && \
#     conda install -y -c bioconda blast=2.11.0 && \
#     conda install bioconda::seqtk && \
#     conda clean -a -y

# RUN mkdir blastdb
# RUN cd blastdb && \
#     wget ftp://ftp.ncbi.nlm.nih.gov/blast/db/16S_ribosomal_RNA.tar.gz && \
#     tar -xvf 16S_ribosomal_RNA.tar.gz && \
#     rm 16S_ribosomal_RNA.tar.gz

# RUN cd blastdb && \
#     wget https://ftp.ncbi.nlm.nih.gov/refseq/TargetedLoci/Bacteria/bacteria.16SrRNA.fna.gz && \
#     gunzip bacteria.16SrRNA.fna.gz && \
#     makeblastdb -in ./bacteria.16SrRNA.fna -dbtype nucl -title 16SMicrobial -out 16SMicrobial -parse_

# cd Docker

# docker build --platform=linux/amd64 -t docker_project:0.1.0 .

# docker run --platform=linux/amd64 -it docker_project:0.1.0 bash

# docker run --platform=linux/amd64 -v "$(pwd)":/data -it docker_project:0.1.0 bash

# cd /data/fastq_pass

# for folder in barcode*/; do
#     for file in "$folder"*.fastq; do
#         seqtk seq -a "$file" > "${file%.fastq}.fasta"
#     done
# done

# cd /data/fastq_pass

# for file in barcode*/*.fasta; do
#     out_file="${file%.fasta}_blast_results.txt"

#     blastn -query "$file" \
#         -db /blastdb/16SMicrobial \
#         -out "$out_file" \
#         -outfmt 6 \
#         -max_target_seqs 3 \
#         -num_threads 8
# done

```

Loading in the data

First, we will load the required packages and classification data

```
library(phyloseq)
library(ggplot2)
library(dplyr)
library(tidyr)
library(purrr)
library(vegan)
library(microbiome)

meta <- read.csv("./metadata.csv")
dat <- read.csv("./stuff.csv")
```

```
names(meta)
str(meta)
names(dat)
str(dat)
```

##Cleaning the data Next, the feature table obtained after decontamination was cleaned and divided into two components: an OTU table and a taxonomy table. The OTU table contains the barcodes(number of DNA sequences) and the tax table links the OTU to its taxonomic classification. Both tables have different data for different analysis

```
otu_dat <- dat[, c(2:14)]
otu_dat <- otu_table(otu_dat, taxa_are_rows = TRUE)
otu_dat <- t(otu_dat)

tax_dat <- dat[, c(16:21, 1)]
tax_dat <- tax_table(tax_dat)
```

```
## Warning in .local(object): Coercing from data.frame class to character matrix
## prior to building taxonomyTable.
## This could introduce artifacts.
## Check your taxonomyTable, or coerce to matrix manually.
```

Now that the OTU and taxonomy tables are created, we must make sure the names are consistent between the two tables. This makes sure that it is easy to read each table.

```
taxa_names(otu_dat) <- rownames(tax_dat)
rownames(meta) <- meta$barcode
```

Lastly, a phyloseq object was constructed to integrate the OTU, taxonomy, and metadata information. We do this so that everything is linked by sample and OTU barcode. This is important for graphing and analysis later on.

```
ps <- phyloseq(otu_dat, tax_dat, sample_data(meta))

colnames(ps@tax_table) <- c("genus", "superkingdom", "kingdom", "phylum", "class", "order", "family")
```

```
head(ps@otu_table)
head(ps@tax_table)
```

Exploratory analysis: Rarefaction curve

In order to see if the sequencing depth is sufficient, we will first create a rarefaction curve of the data. This will show how the number of observed OTUs changes with increasing sequencing depth. This ensures that we collect a sufficient amount unique samples so that the total diversity is reasonable. We conduct the rarefaction analysis by sampling the taxa at different sequencing depths and plotting the number of observed OTUs. In this case we will sample the data at intervals of 1 read and remove any taxonomic unit with less than 10 reads.

```
ps.rare <- prune_taxa(taxa_sums(ps) > 10, ps)
rare_table <- otu_table(ps.rare)
rare_table <- as.data.frame(otu_table(ps.rare))
sample_names <- rownames(rare_table)
raredat <- rarecurve(rare_table, step = 1, label = TRUE, tidy = TRUE)
```

```
## Warning in rarecurve(rare_table, step = 1, label = TRUE, tidy = TRUE): most
## observed count data have counts 1, but smallest count is 18
```

```
rareplot <- ggplot(raredat, aes(x = Sample, y = Species, color =
Site)) +
labs(caption = "Figure 1: Rarefaction of Sampled Bacteria Species")+
geom_point()
```

```
rareplot
```

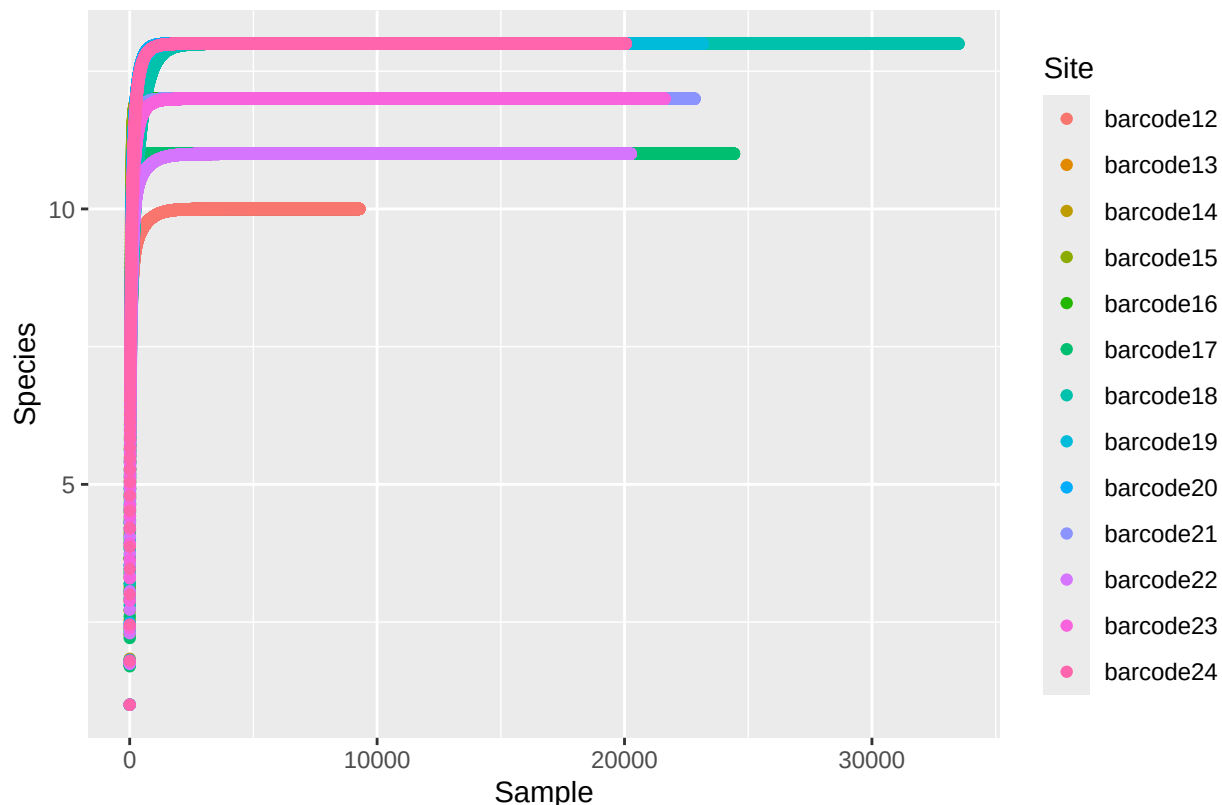


Figure 1: Rarefaction of Sampled Bacteria Species

The rarefaction curve shows that the sequencing depth is enough for majority of the samples. As the number of barcodes from the OTU begin to plateau, that means that most of the unique species have been gathered, making total diversity as accurate as possible. ## Exploratory analysis: Relative abundance We will now conduct an exploratory analysis of the data to determine the relative abundance of the bacterial taxa in the samples. To clean up the data, we will first remove any taxa contain <100 reads since they are regarded as “noise.” This can skew data and make patterns harder to detect. We will first calculate the relative abundance of the taxa in the samples. Since different samples have different sequencing depths. Calculating the relative abundance will allow us to compare microbiome composition proportionally. We will then plot the relative abundance of the taxa in the samples to visually see the abundance of species in the humans intestines.

```
ps.prune <- prune_taxa(taxa_sums(ps) > 100, ps)
cfam <- tax_glom(ps.prune, taxrank = rank_names(ps.prune)[5])
cfam <- transform_sample_counts(cfam, function(x) x / sum(x) * 100)
```

```
melt <- psmelt(cfam)
melt <- melt %>%
  group_by(barcode, timepoint, class) %>%
  summarize(RelativeAbundance = sum(Abundance))
```

```
## 'summarise()' has grouped output by 'barcode', 'timepoint'. You can override
## using the '.groups' argument.
```

```
plot <- ggplot(melt, aes(x = barcode, y = RelativeAbundance, fill = class)) +
  geom_bar(stat = 'identity', position = 'stack') +
  labs(caption = "Figure 2: Relative Abundance of Gut Bacteria in each ID ")
```

```
plot
```

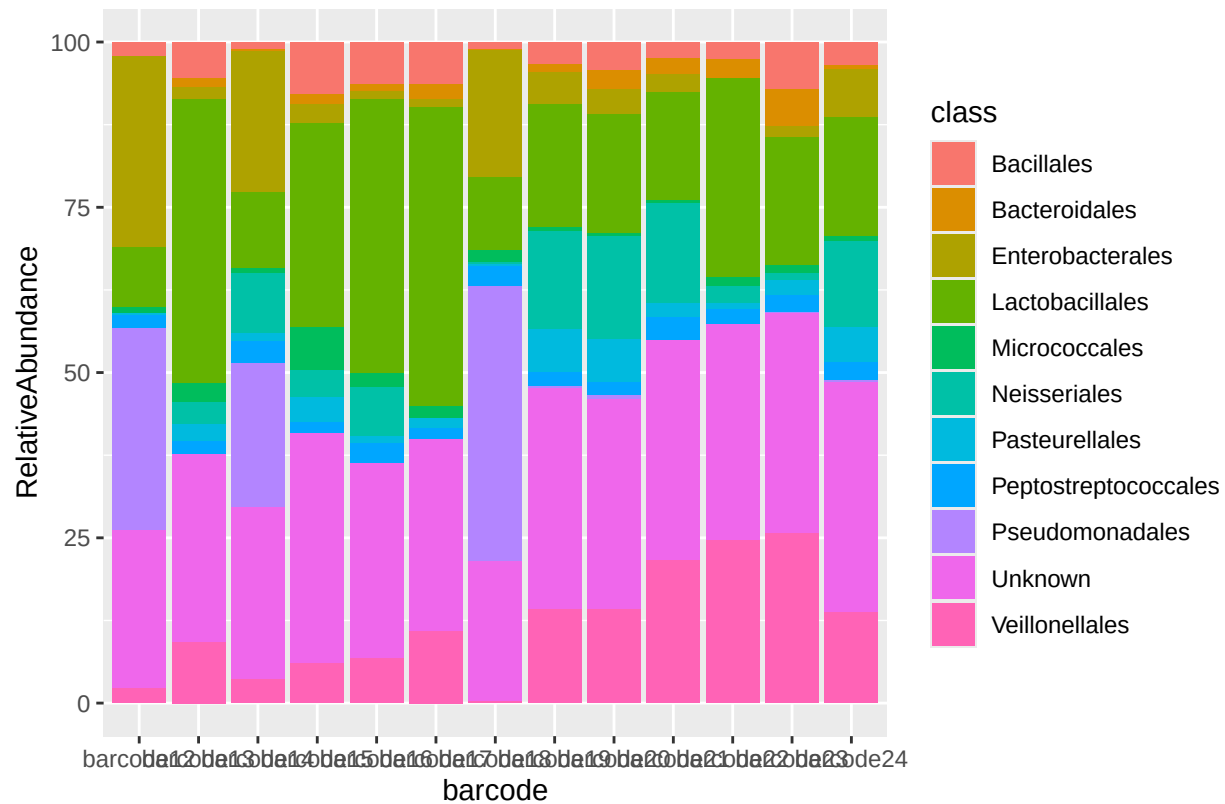


Figure 2: Relative Abundance of Gut Bacteria in each ID

From looking at this relative abundance barplot, we can see that barcodes 19-24 had a higher relative abundance of Veillonellales (pink) and Neisseriales, while it decreased the relative abundance of Lactobacillales (dark green) and Pseudomonadales (purple). To get further evidence that Red 40 synthetic food dye really affects human gut microbiome, we will need to conduct a more rigorous analysis to determine if the samples are actually different from each other.

Alpha diversity analysis

In order to quantify the diversity within the samples, we will calculate the alpha diversity of each sample. We will find the observed species and Simpson diversity indices for the samples. The observed species index measures the number of unique species present in the samples, while the Simpson index indicates the diversity of the sample. We will then plot the alpha diversity indices for the samples. Using both the observed species index and Simpson, we can create a hypothesis that lower diversity means synthetic food does decrease human gut microbiome since it produces more of a specific enzyme to break down Red 40.

```
alpha_indiv1 <- plot_richness(ps, x = "samples", measures = "Observed")+
  labs(caption = "Figure 3: Observed species index of Samples")+
  geom_boxplot(aes(color = samples))
```

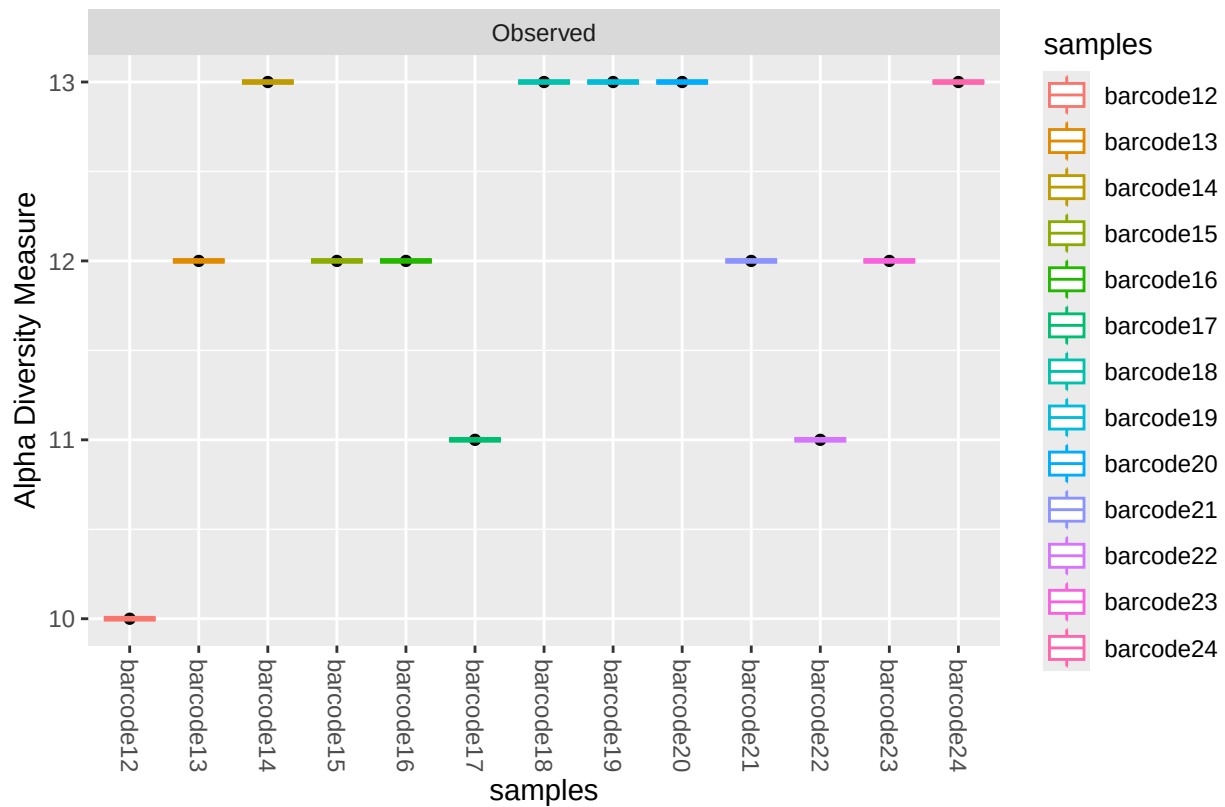
```
## Warning in estimate_richness(physeq, split = TRUE, measures = measures): The data you have provided contains
## any singletons. This is highly suspicious. Results of richness
```

```
## estimates (for example) are probably unreliable, or wrong, if you have already
## trimmed low-abundance taxa from the data.
##
## We recommended that you find the un-trimmed data and retry.
```

```
alpha_indiv2 <- plot_richness(ps, x = "samples", measures = "Simpson")+
labs(caption = "Figure 4: Simpson index of Samples") +
geom_boxplot(aes(color = samples))
```

```
## Warning in estimate_richness(physeq, split = TRUE, measures = measures): The data you have provided
## any singletons. This is highly suspicious. Results of richness
## estimates (for example) are probably unreliable, or wrong, if you have already
## trimmed low-abundance taxa from the data.
##
## We recommended that you find the un-trimmed data and retry.
```

```
alpha_indiv1
```



```
alpha_indiv2
```

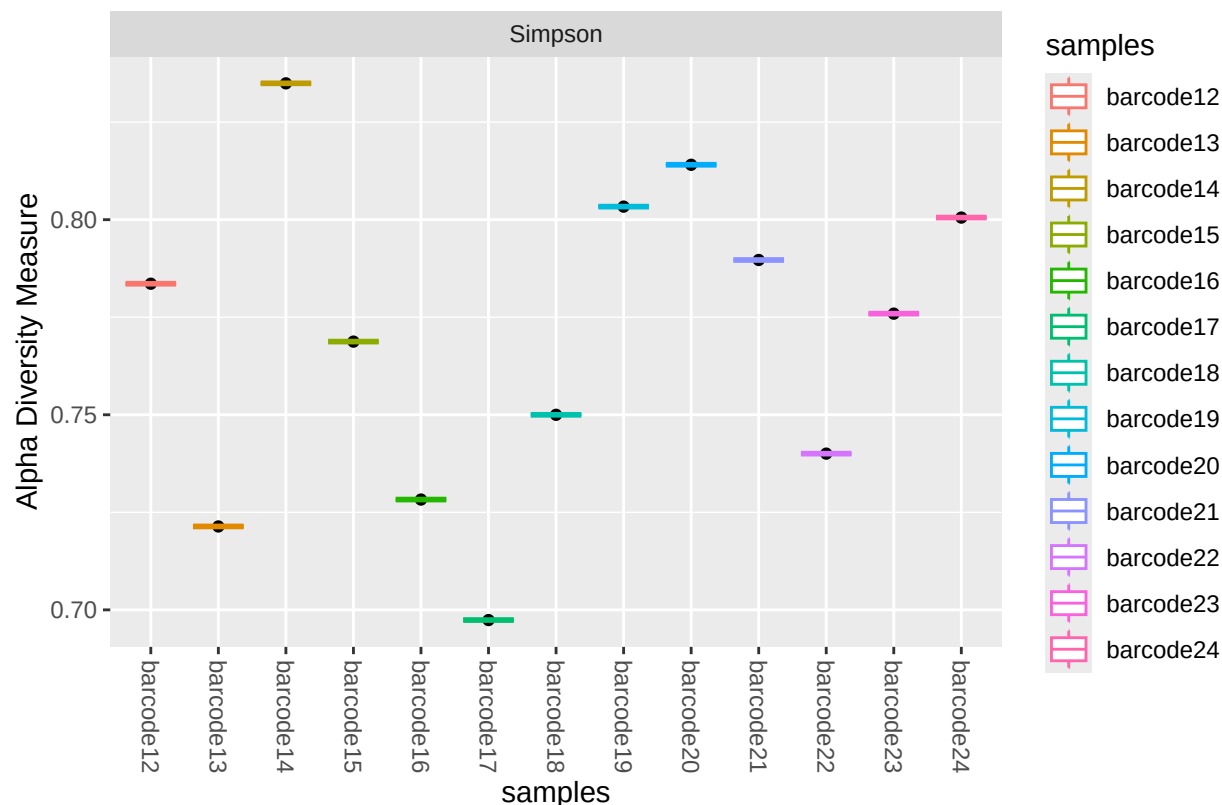


Figure 4: Simpson index of Samples

When looking at the Observed index, a higher index indicates a high amount of diversity. The Simpson index tells us how evenly distributed the sample is. For the observed index, we can see a cluster of samples where there are 13 unique species and another cluster where there are 12 unique species. This tells us that each sample has high amounts of unique species with the amount of unique species hovering around 12 and 13. When taking a look at the Simpson index of samples graph, we can see that the diversity of bacteria species vary between all the barcodes. You can argue that there is a cluster in the top right where there is a higher diversity. With the observed index and Simpson index taken into account, the barcodes after the Red 40 was consumed had a high alpha diversity.

Explanatory Analysis: PCoA

We will now conduct a principle coordinate analysis (PCoA) of the samples to detect similarities in bacterial composition at the genus/species level. This is important because it provides further proof of whether consuming synthetic food dye contributes to the diversity of bacteria in the human gut. We will first convert the data to compositional data and then create a distance matrix based on Bray-Curtis dissimilarity. The distance matrix will then be used to conduct the PCoA. The results of the PCoA will be plotted.

```
sample_variables(ps)
```

```
## [1] "barcode" "id" "timepoint"
```

```
ord <- ordinate(ps, method="MDS")
pcoa_plot <- plot_ordination(ps, ord, color="barcode")+
labs(caption = "Figure 5: pCoA plot of bacteria species")
```

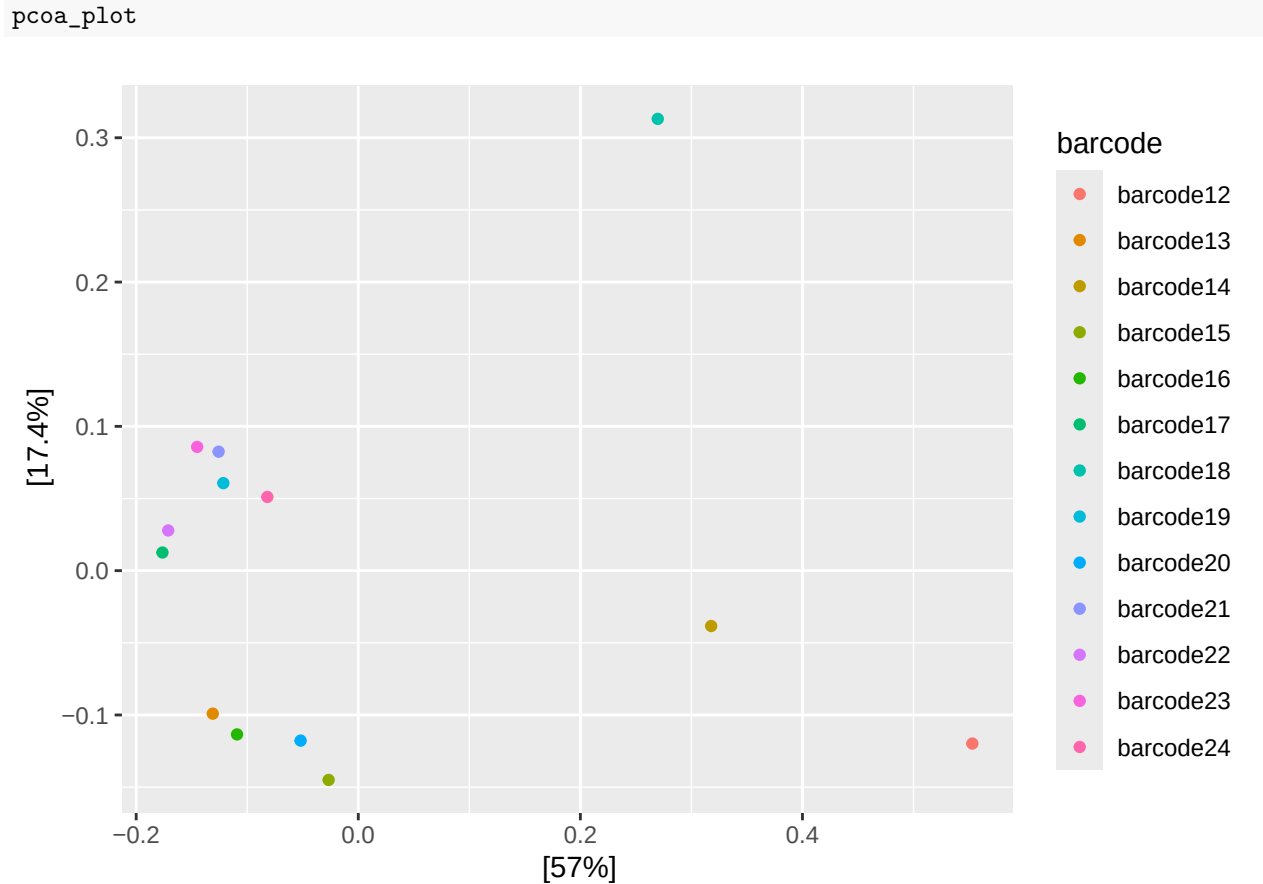


Figure 5: pCoA plot of bacteria species

From the PCoA plot, we can see a cluster of dots in the middle right and another cluster in the bottom right. Just by looking at the colors of the dots, the most of the dots in the middle right are between 19-24 while the bottom right contains barcodes 12-18. This indicates that for barcodes 12-18, they have similar bacteria diversity and barcodes 19-24 have similar bacteria diversity.

Conclusion

From the analysis, we can conclude that the consumption of Red 40 or synthetic food dye contributes to a higher diversity of bacteria in the human gut microbiome. Through the pre-processing stage, we determined that barcodes 12-18 were baseline(pre-Red40 consumption) while barcodes 19-24 were after Red40 consumption. Through the refraction curve, we know that the sampling depth for all samples is sufficient. With the relative abundance graph of each barcode, the consumption of Red 40 added more abundance of a certain speicies while decreasing another when comparing barcodes between baseline and Red40. There is more convincing evidence of higher diveristy when we examined the observed species index and the simpson index of the samples. Based on the figures, there was a high amount of obsered unique species from all barcodes ranigng from 12-13. When looking at the simpson index, we found that barcodes 19-24 clustering at the top symbolzing a higher diversity among the gut microbiome. To furthur explain this phenomenon, we graphed the PCoa graph to take a deeper look of the similarity of genus and class in each barcode. We confirmed that barcodes 19-24 and barcodes 12-18 had similar gut microbiome makeup. We can therefore conclude that consuming Red40 will produce more unique bacteria in the human gut while remaining relatively diverse amongst the bacteria before Red 40 was consumed. This could've happened because the intestine created more enzymes to break down the synthetic food dye consumed. As a result, there were more unique bacteria in the human gut microbiome without completely dominating the bacteria population before Red 40 was

consumed. This study can provide key reserach for scientists looking to determine if synthetic food dye is an acutal threat to human health. A deeper look into the type of bacteria being produced by syntheitc food dye could generate more interesting findings.

```
library(rmarkdown) library(tinytex) render("sigma.Rmd", output_format = "pdf_document")
```